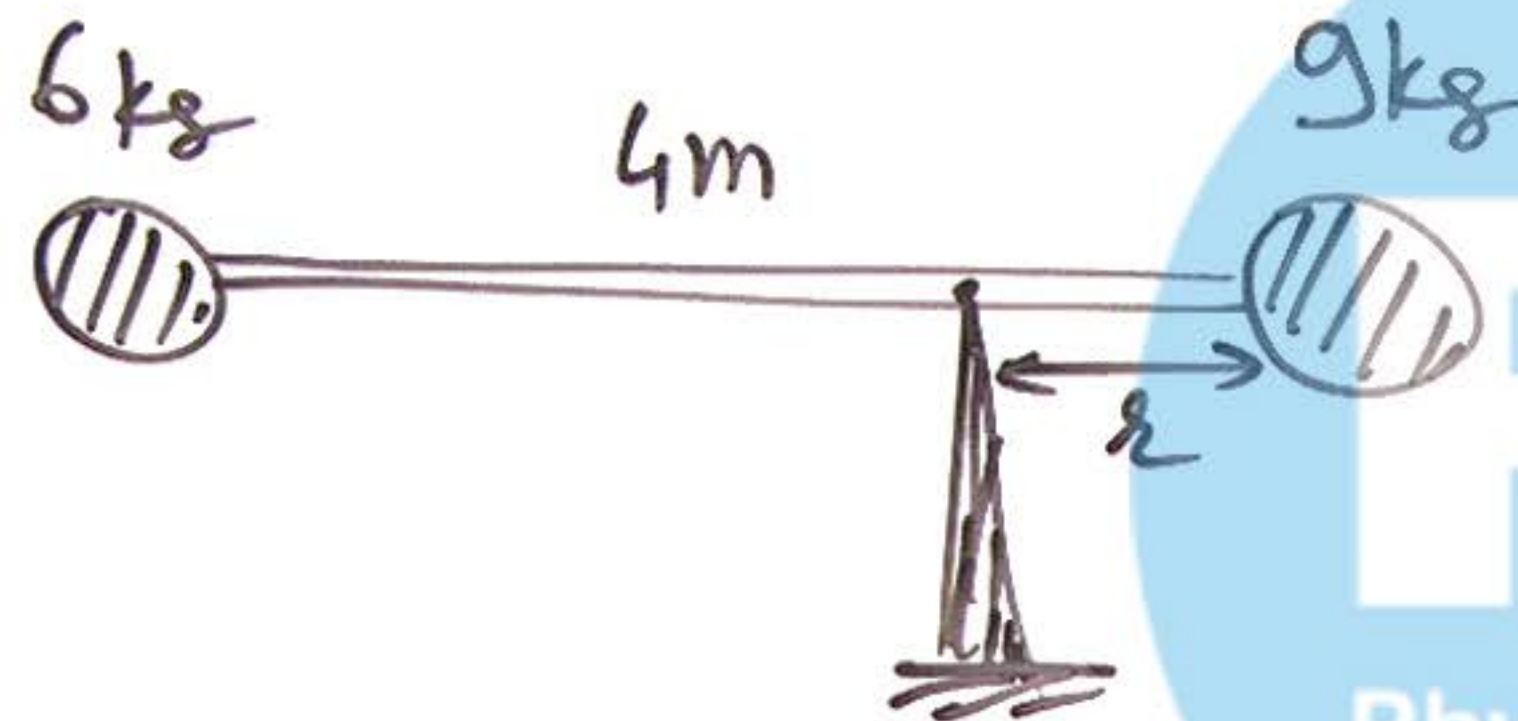
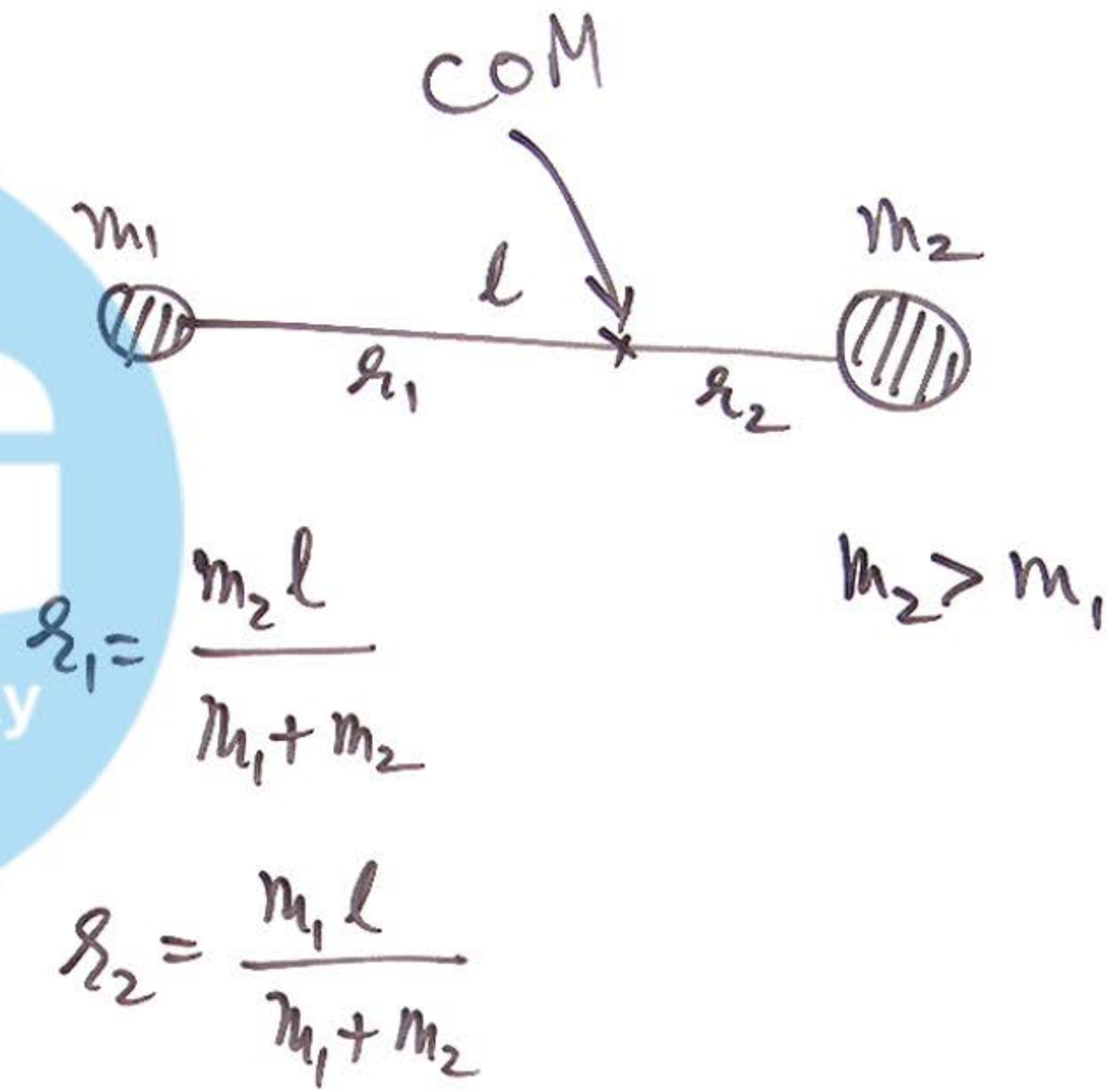


① COM of a 2-Body System:

Ex:

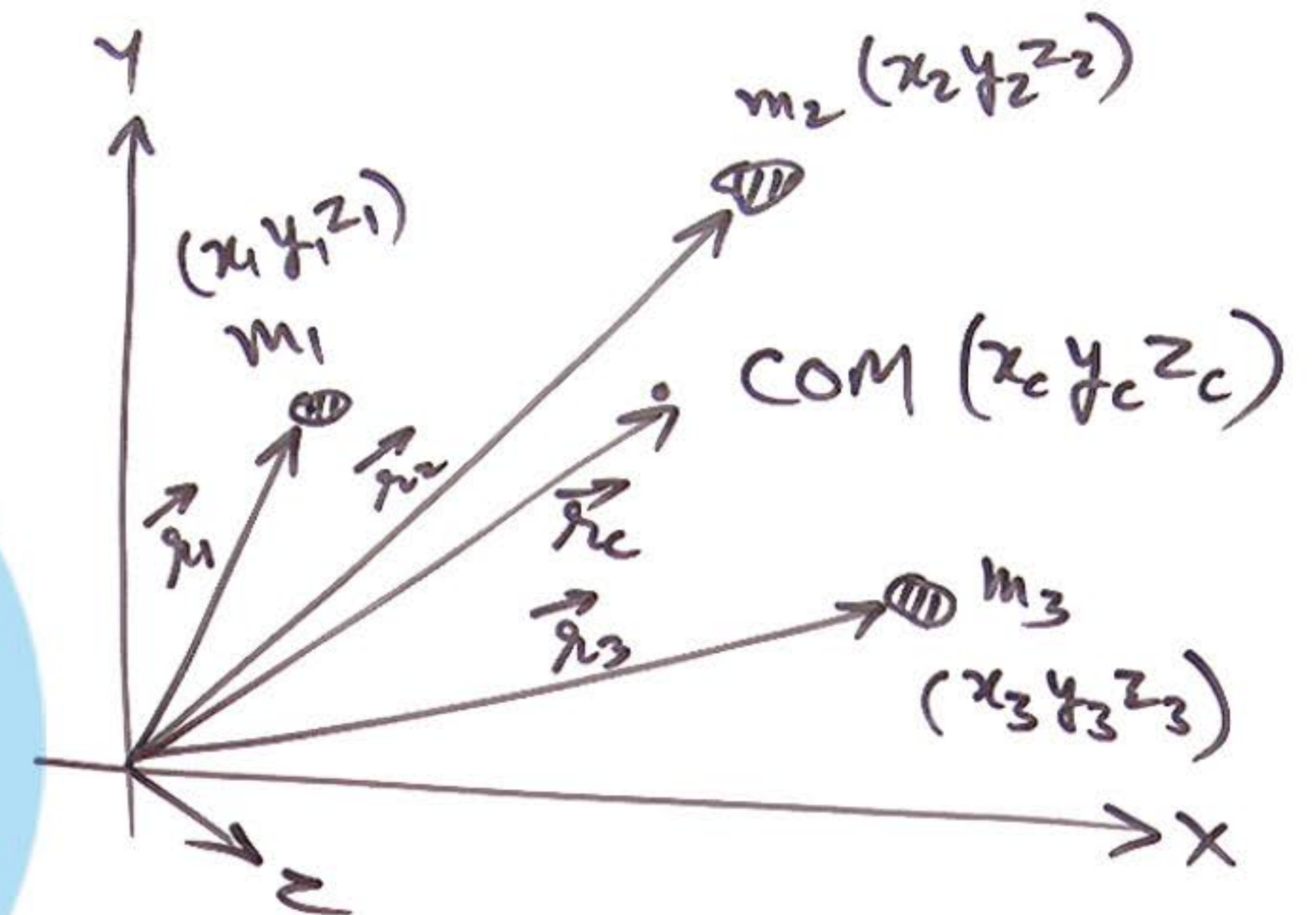


$$x = \frac{6 \times 4}{15} = \frac{24}{15} \text{ m}$$



② COM of a multiparticle System:

$$\vec{r}_c = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3}{m_1 + m_2 + m_3}$$



$$x_{cm} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3}$$

$$y_{cm} = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3}{m_1 + m_2 + m_3}$$

$$z_{cm} = \frac{m_1 z_1 + m_2 z_2 + m_3 z_3}{m_1 + m_2 + m_3}$$

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③ Concept of "mass moment" in COM

vector $\vec{L}$

$$\vec{L} = m \vec{r}$$

mass of
particle

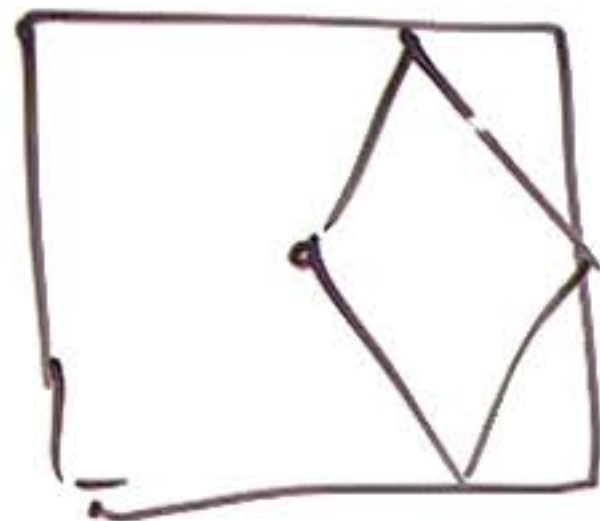
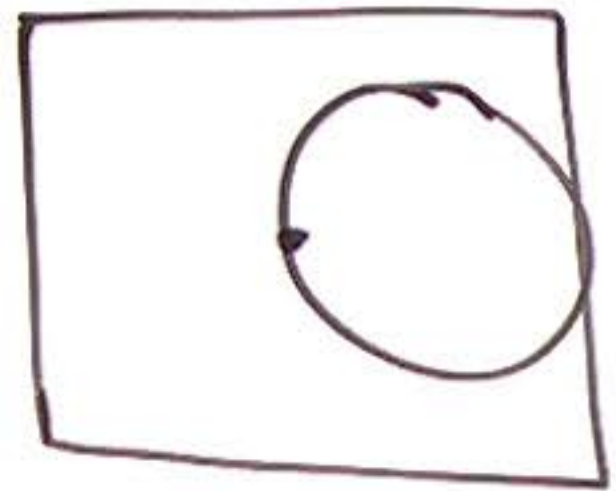
p.v. of particle

About COM sum of mm
of all particles is
always zero.

$$\vec{r}_{\text{com}} = \frac{\sum_{i=1}^n m_i \vec{r}_i}{\sum_{i=1}^n m_i}$$

④ COM of Subtractive bodies:

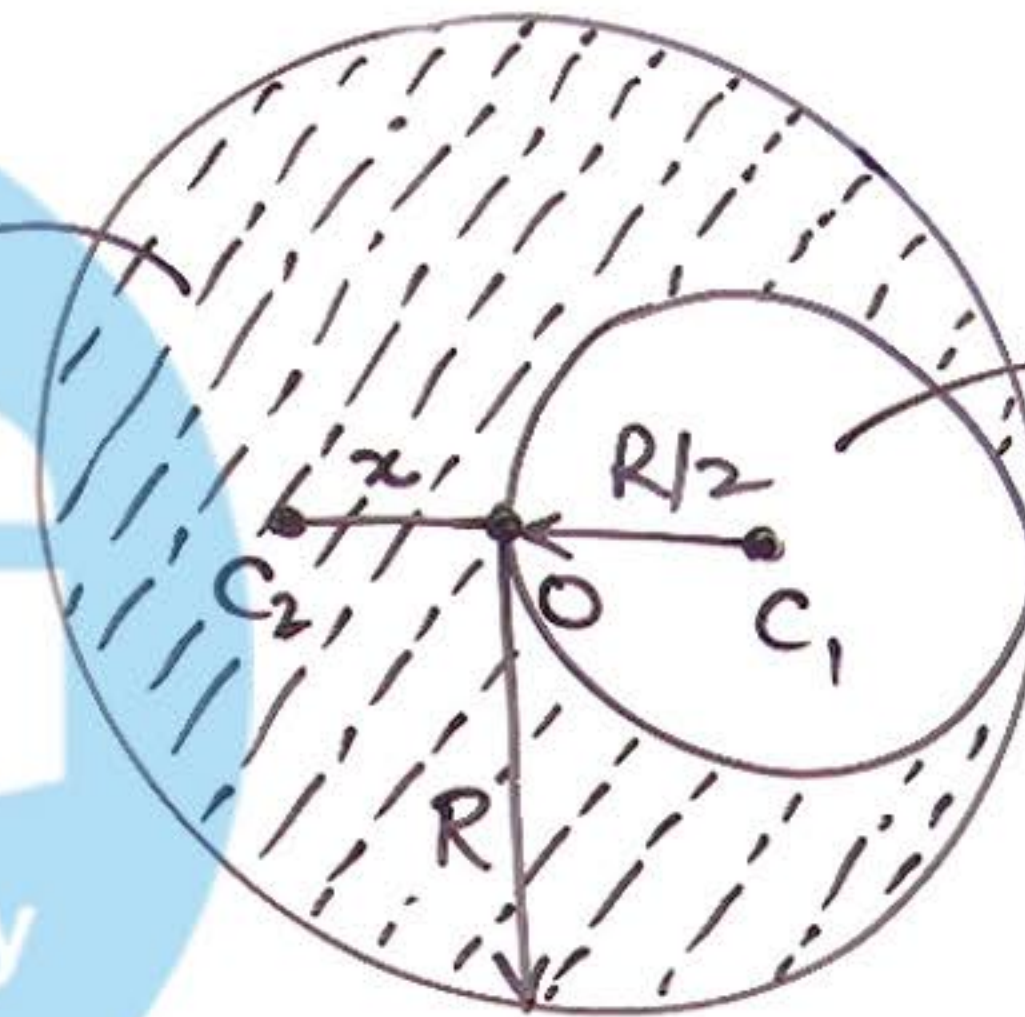
Using prop of mm, we use



$$m_0 \times R/2 = 3m_0 \times x$$

$$\Rightarrow x = R/6 \text{ Ans}$$

$$m_2 = C \times \frac{3\pi R^2}{4} = 3m_0$$

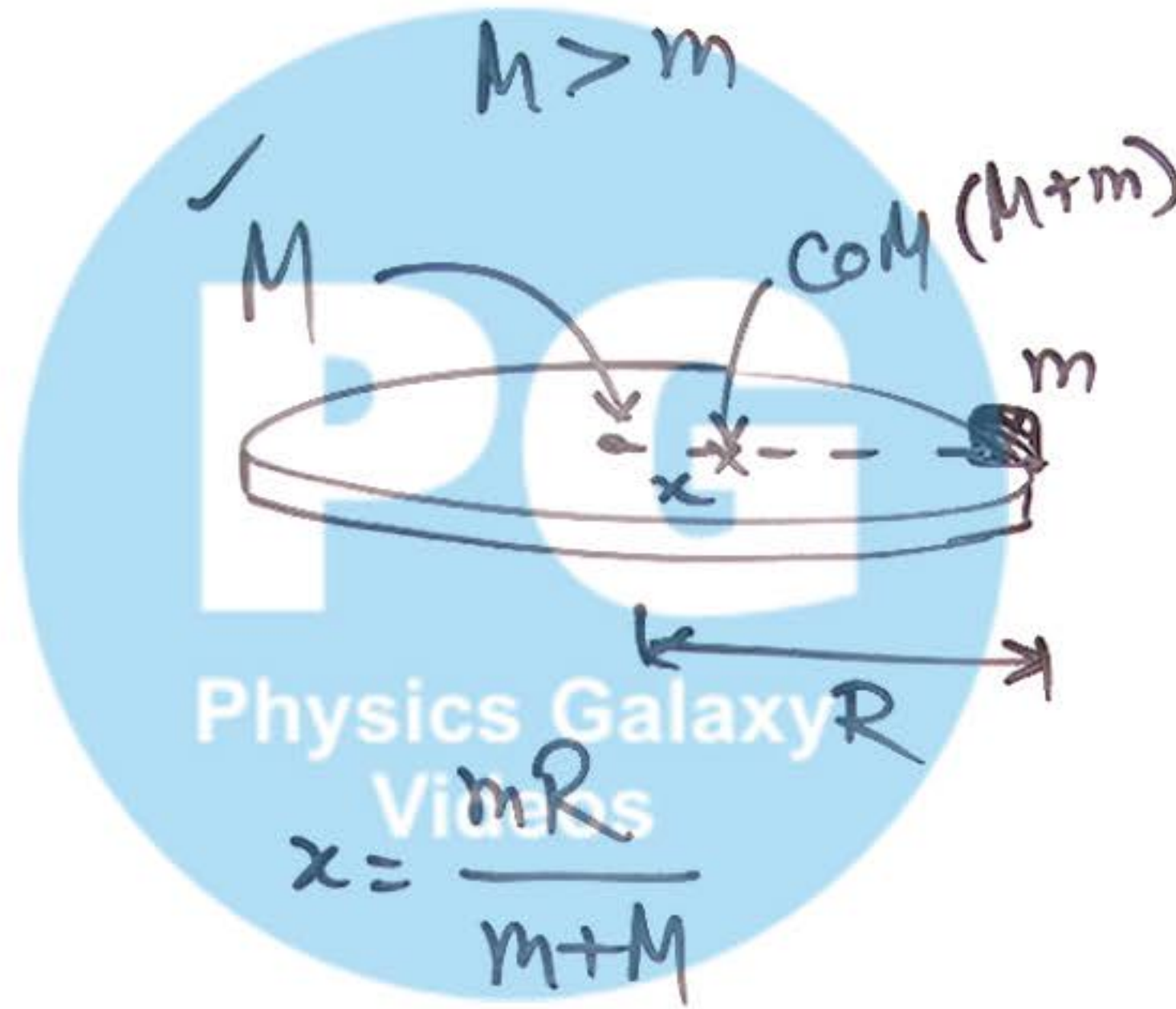


$$m_1 = C \times \frac{\pi R^2}{4} = m_0$$

in case of sphere

$$m_1 = C \times \frac{4}{3} \pi (R/2)^3$$

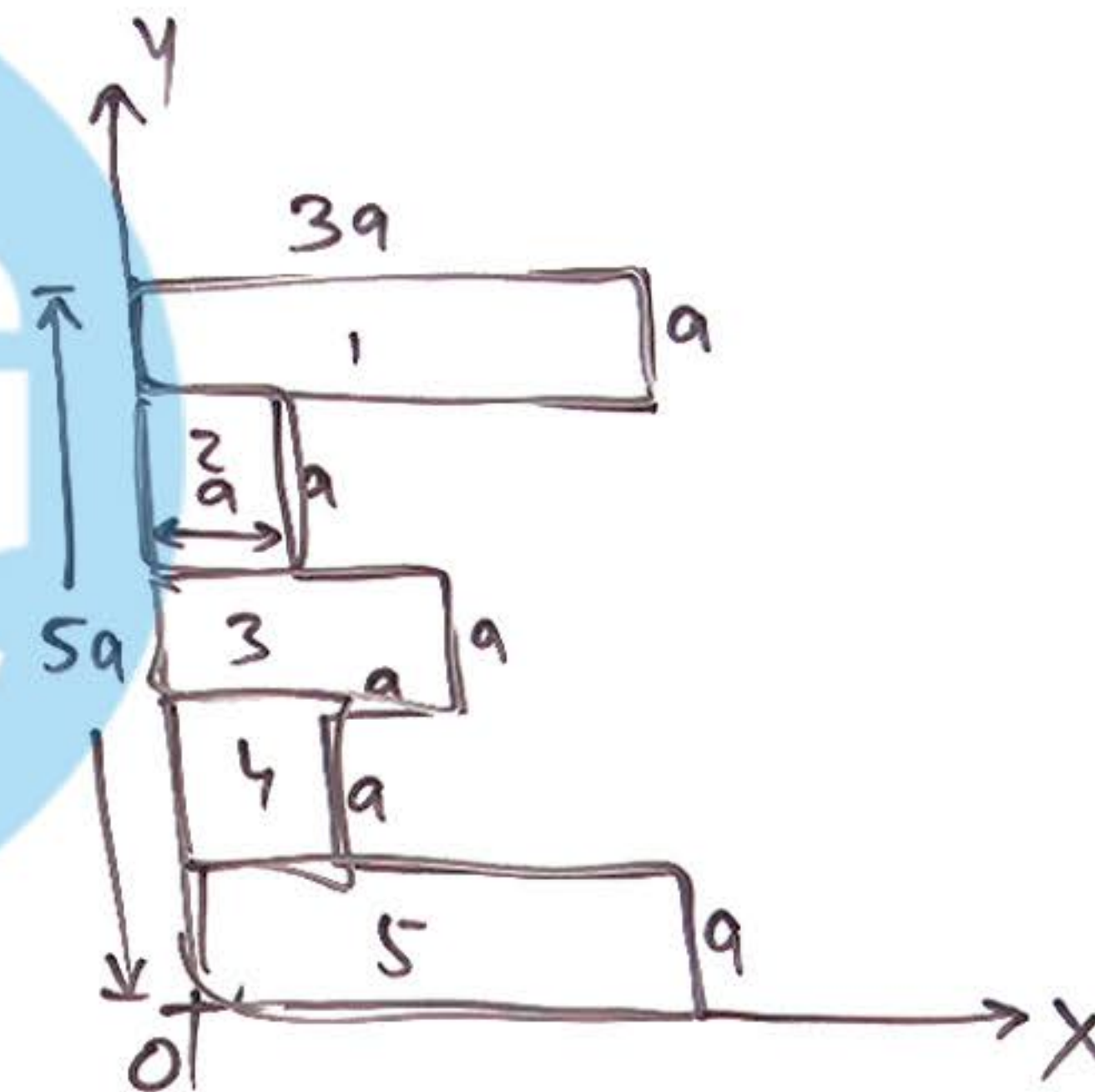
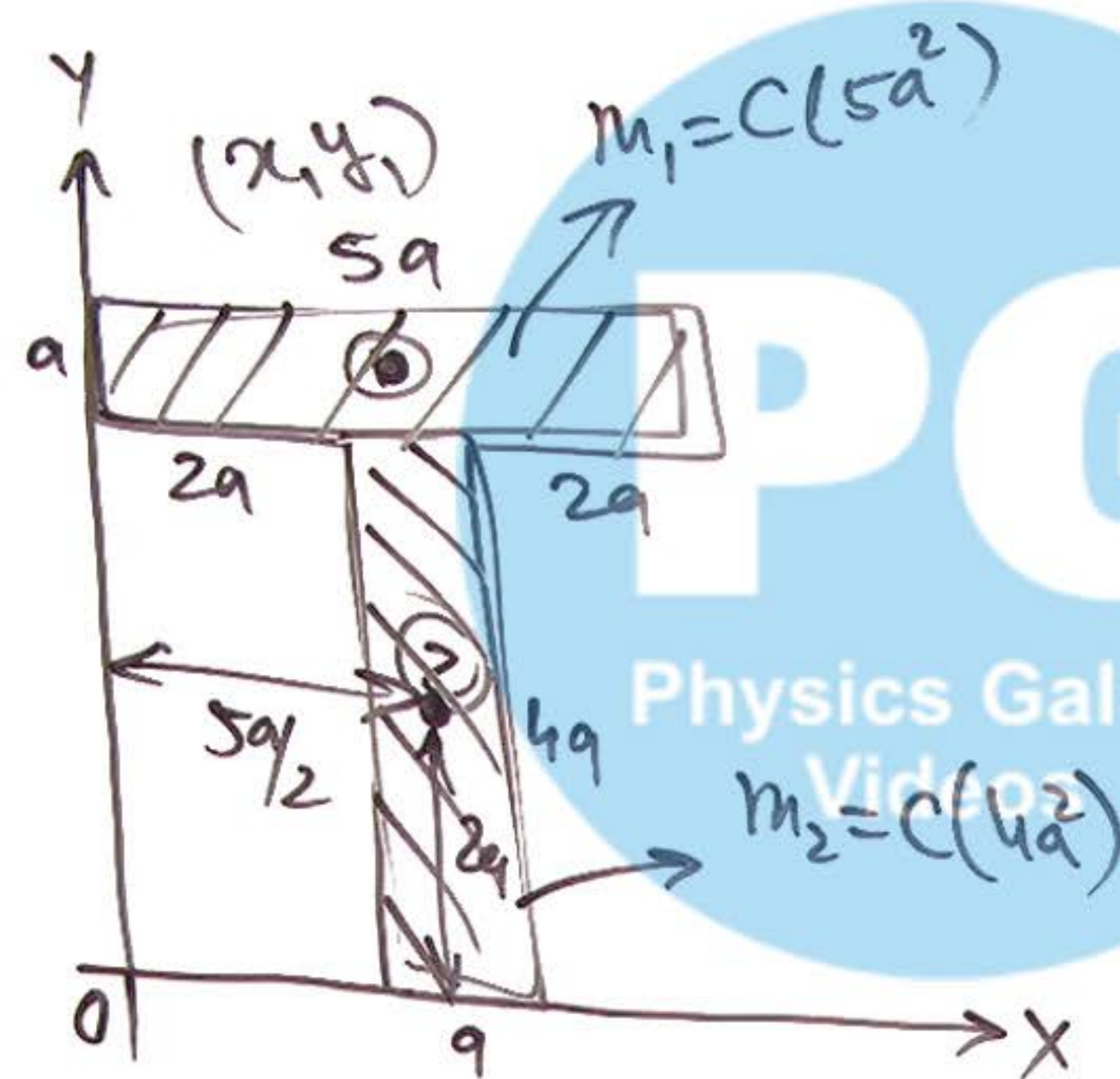
⑤ COM of Additive Objects :



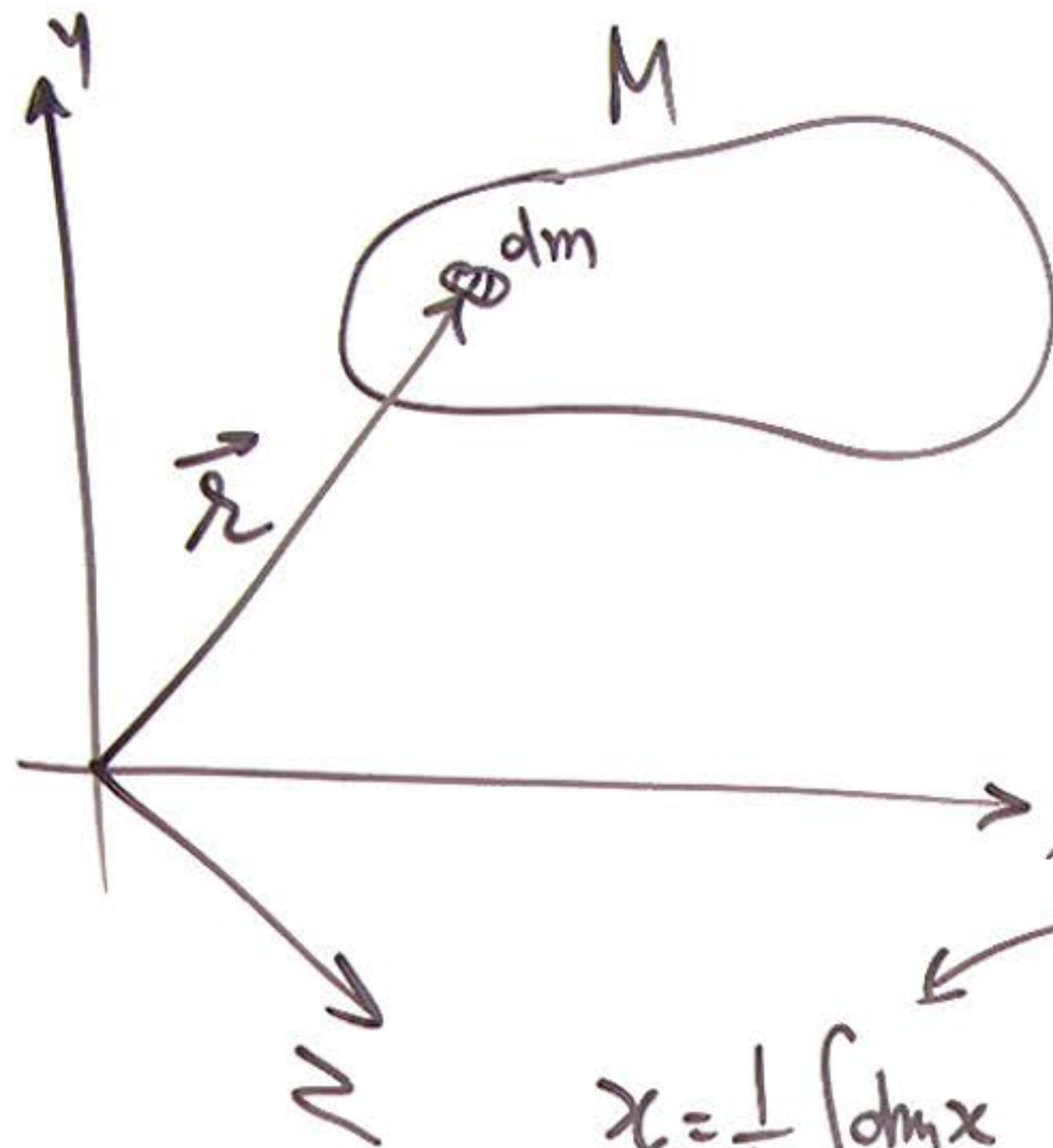
⑥ COM of flat geometrical figures:

$$x_c = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$y_c = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$



⑦ COM of a Continuous Extended Body :



$$\vec{r}_c = \frac{\int dm \vec{r}}{\int dm}$$

$$\vec{r}_c = \frac{1}{M} \int dm \vec{r}$$

$$x_c = \frac{1}{M} \int dm x$$

$$y_c = \frac{1}{M} \int dm y$$

$$z_c = \frac{1}{M} \int dm z$$



$$y_c = \frac{2R}{\pi}$$



$$y_c = \frac{4R}{3\pi}$$



$$y_c = \frac{R}{2}$$



$$y_c = \frac{3R}{8}$$

⑧ COM of Composite Objects:

II: find distance of COM from O.

dist of com from O

$$y_0 = \frac{m_1 \left(-\frac{3R}{8} \right) + m_2 \left(\frac{h}{4} \right)}{m_1 + m_2}$$

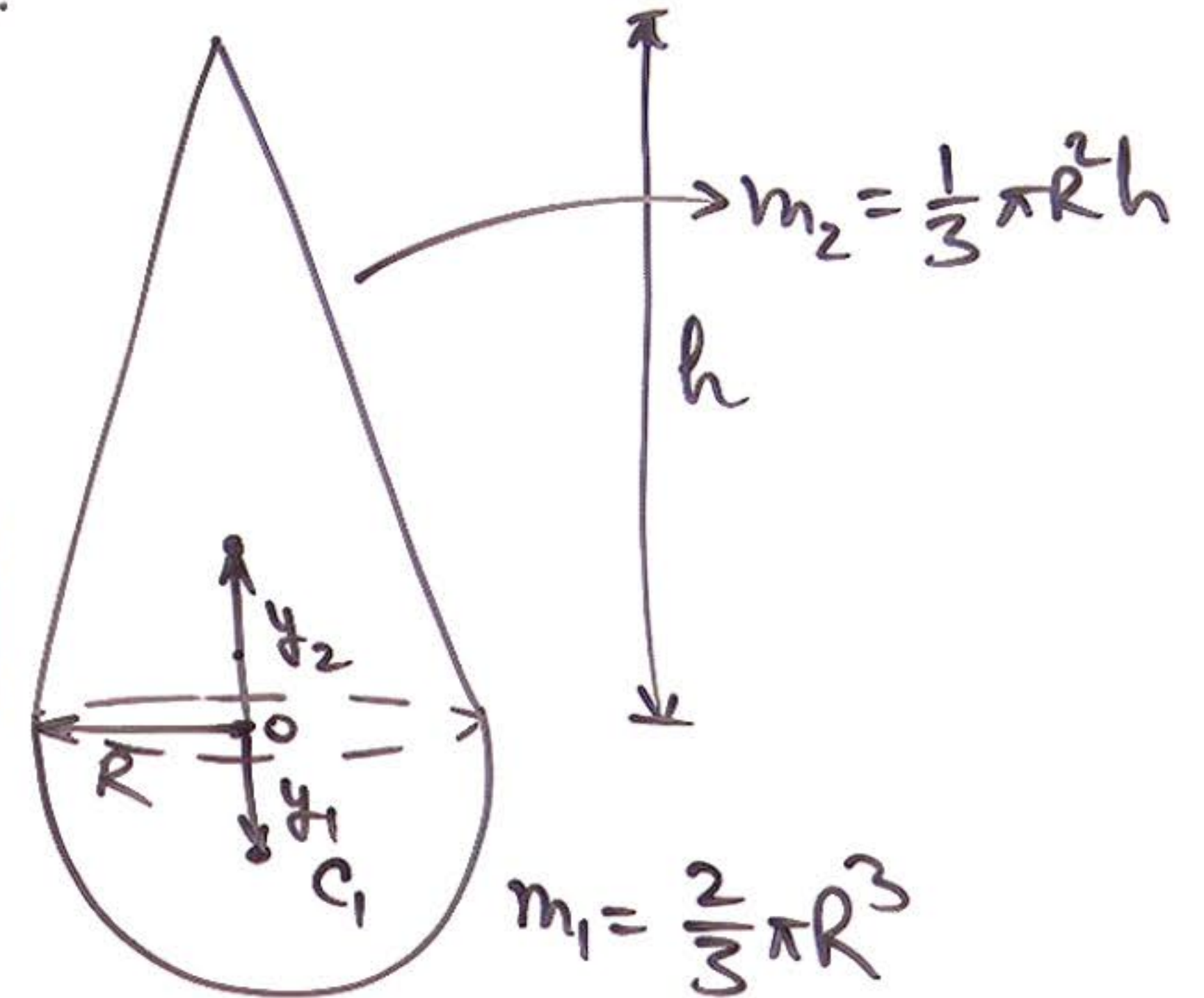
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$$y_2 = \frac{h}{4}$$

$$y_1 = \frac{3R}{8}$$



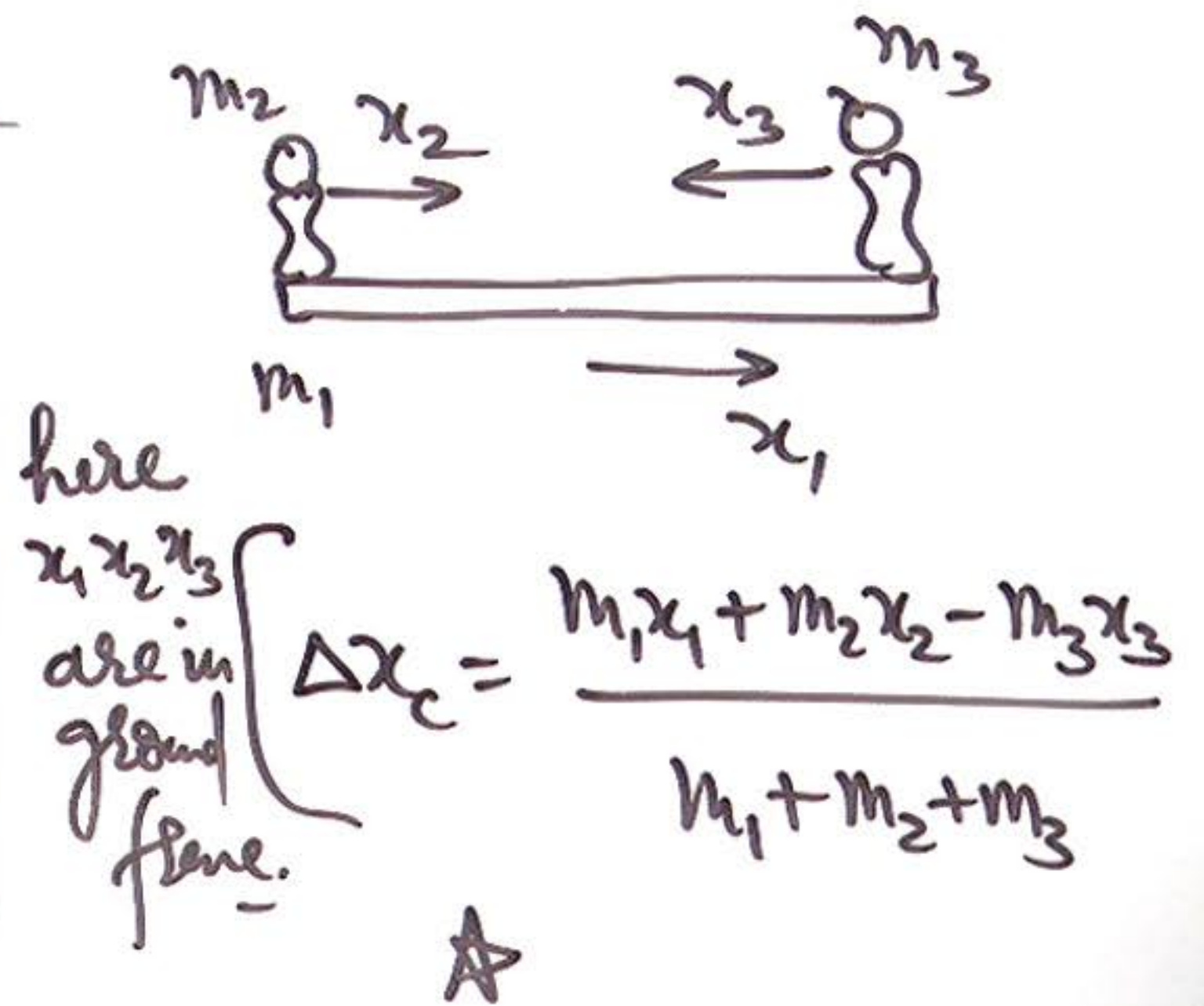
⑨ Displacement of CoM of a System:

* if x_2 & x_3 are meas
on plank then
we use.

$$\Delta x_c = \frac{m_1 x_1 + m_2 (x_2 + x_1) - m_3 (x_3 - x_1)}{m_1 + m_2 + m_3} \Rightarrow$$

$$\vec{r}_c = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + \dots}{m_1 + m_2 + \dots}$$

$$\Delta \vec{r}_c = \frac{m_1 \Delta \vec{r}_1 + m_2 \Delta \vec{r}_2 + \dots}{m_1 + m_2 + \dots}$$



⑩ Velocity & Acceleration of CoM:

If on a System $\vec{F}_{\text{ext}} = 0$
(Net)

$$\Rightarrow \vec{a}_c = 0$$

$$\Rightarrow \vec{V}_c = \text{Constant}$$

$$\Rightarrow \vec{P}_{\text{System}} = \text{Constant}$$

$$\vec{r}_c = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3 + \dots}{m_1 + m_2 + m_3 + \dots}$$

$$\vec{V}_c = \frac{d\vec{r}_c}{dt} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2 + \dots}{m_1 + m_2 + \dots} = \frac{\vec{P}_{\text{System}}}{\sum_{i=1}^n m_i}$$

$$\vec{a}_c = \frac{d\vec{V}_c}{dt} = \frac{m_1 \vec{a}_1 + m_2 \vec{a}_2 + \dots}{m_1 + m_2 + \dots} = \frac{\sum \vec{F}_{\text{ext}}}{\sum_{i=1}^n m_i}$$